

# **chimerax-vampnet: Interactive Markov state modeling of protein conformational landscapes from heterogeneous ensembles, with a neural-organoid-relevant demonstration on Notch1 NRR**

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## **Abstract**

Protein conformational landscapes are now accessible from three distinct computational sources: classical molecular dynamics (MD), flow-matching generative ensembles (ALPHAFLOW [12]), and neural Boltzmann samplers (BIOEMU [15]). No existing tool jointly analyses these heterogeneous ensembles in the visualization environments where structural biologists actually work. We present **chimerax-vampnet** (v0.1), a UCSF ChimeraX bundle that fits a single VAMPnet [17] to the union of MD frames, AlphaFlow samples, and BioEmu samples, providing per-state identification, slow-mode visualization, and a Markov state model (MSM) [10] in the same interactive 3D context as the structures themselves. Every command returns JSON-serializable data and is exposed via a Model Context Protocol [1] bridge, enabling an LLM agent (Claude Desktop, Cursor) to drive an adaptive sampling pipeline that extends the ensemble until the inferred state space converges. We validate the bundle as a unit test on chignolin (CLN025) — a 1  $\mu$ s explicit-solvent trajectory generated on Modal H100 GPUs — recovering folded  $\leftrightarrow$  unfolded state separation and a slowest implied timescale of 166 ns. We further establish bundle robustness on four ATLAS public MD trajectories [21] spanning all- $\alpha$ , mostly- $\beta$ , mixed, and large- $\alpha$  folds (73–518 residues): all four converge to non-degenerate 4-state decompositions with slowest timescales of 38–75 ns in under two minutes of wall-clock each. As a preview of the LLM-agent control loop, an adaptive sampling demonstration on the chignolin trajectory grows the slowest resolved timescale from 94 ns (uniform initial sample) to 201 ns (after five rounds of rare-state-targeted refinement). We apply the bundle to the Notch1 Negative Regulatory Region (NRR) — the conformational switch that gates neural-stem-cell fate decisions in cerebral organoids [11, 14]. The apo (3I08) and corrected-chain holo (3L95 chains X+H+L, NRR + Fab pair) trajectories ( $3 \times 100$  ns each on Modal A100-80GB) are complete. The v0.2 H2 analysis finds the directional prediction met: apo is +3.7 percentage points more auto-inhibited than holo (21.6 % vs 17.9 %), matching the published mechanism [7, 20, 24]. The pre-registered magnitudes (apo  $\geq 50$  %, holo  $\leq 30$  %) are *not* met because both systems undergo NRR-fragment dissociation across the 100 ns trajectory in the absence of a working transmembrane-anchor restraint; recovering the magnitudes is the first v0.3 task. The bundle (1184 LOC, 18/18 tests passing) is released under MIT at <https://github.com/m9h/chimerax-vampnet>.

**Status (v0.4):** bundle shipped, four GitHub releases tagged (v0.1, v0.1.1, v0.2, v0.3); chignolin Tier-1 validation passes; four-protein ATLAS robustness sweep passes; LLM-agent adaptive sampling demonstration passes (v0.1 proxy) plus a live MCP-driven adaptive sampling end-to-end PoC (v0.4) that triggers a real Modal MD launch on agent decision; Notch1 NRR apo + corrected-chain holo MD complete under two restraint protocols; H2 directional prediction met (+3.7/+3.8 pp under both protocols) but pre-registered magnitudes blocked by the 100 ns sampling horizon. v0.4 adds MarS-FM and BioEmu as additional ensemble sources, with **\*\*H3** multi-source coverage MET<sup>\*\*</sup>: state 2 of the joint VAMPnet (9 % of the joint ensemble) is reached only by generative samples (96.4 % from BioEmu, 3 % from MarS-FM, zero from MD) — direct evidence that generative ensembles surface conformations classical 100 ns MD does not reach. Bootstrap CIs on H2 indicate the +3.8 pp directional Delta is not statistically significant at  $n = 3$  replicas (95 % CIs overlap). AlphaFlow and Boltz-2 integration deferred to v0.5 (upstream-dep-stack incompatibilities; neither model is on the H3-critical path).

## 1 Introduction

The question “*what conformational states does this protein populate?*” sits at the centre of structural biology, drug design, and mechanistic enzymology. Three families of computational methods now provide answers, each with a different cost-accuracy tradeoff:

1. **Molecular dynamics (MD)** on a force field (AMBER, CHARMM, OPLS) is physically rigorous but slow. Even with modern GPU engines [5], sampling beyond microseconds requires specialized hardware or enhanced sampling.
2. **Flow-matching / diffusion generative models** [12] learn to sample protein conformations directly from sequence or structure, producing diverse ensembles in minutes but without explicit dynamical information.
3. **Neural Boltzmann samplers** [15] train on MD ensembles to draw approximate equilibrium samples without running new MD — effectively an amortised, fast surrogate for MD’s stationary distribution.

Each of these sources gives a different cut at the same underlying landscape, and combining them strictly dominates any single source: MD provides ground-truth kinetics, AlphaFlow provides rapid breadth, BioEmu provides cheap density estimation. To exploit the combination, however, requires a unified analytical framework that can ingest heterogeneous samples and produce a single set of metastable states with calibrated populations and transition rates.

Variational approach for Markov processes (VAMP) networks [17, 23] provide exactly that framework. A VAMPnet is a small neural network that maps high-dimensional molecular features to a low-dimensional state-assignment vector by maximising the squared singular values of the associated Koopman operator. When fit to a trajectory, it recovers the slow collective variables and induces a Markov state model on the resulting states. The methodology is structurally identical to the hidden-Markov / dynamic mode decomposition approaches that the neuroimaging community has developed for MEG/EEG state inference [6, 22]; the cross-modal opportunity is that the brain-state methodology can be applied verbatim to molecular dynamics, and a practitioner fluent in one is well-positioned to bridge to the other.

Existing implementations of VAMPnets (DEEPTIME, the successor of PYEMMA [8]) produce numpy arrays and matplotlib plots. They are decoupled from the 3D visualization environments

— UCSF ChimeraX [19] and PyMOL — where structural biologists interactively explore conformations. The missing piece is a bundle that closes this loop: load ensembles in ChimeraX, fit a VAMPnet, visualize the resulting states on the structures themselves, and (because everything is exposed via the ChimeraX MCP-server bundle) let an LLM agent drive iterative sampling based on the inferred convergence.

We further demonstrate the bundle on a specific molecule of broad biological interest: the Notch1 Negative Regulatory Region (NRR), a 350-residue conformational switch whose autoinhibited and activated conformations are well-characterised crystallographically [7, 20]. Notch1 is the master regulator of neural-stem-cell self-renewal vs. differentiation in cerebral organoid models of human brain development [2, 11, 14]. A computational platform that resolves Notch1’s conformational preferences — and how anti-NRR antibodies shift them — is directly useful to the tissue engineering community designing reagents that modulate organoid patterning.

## 2 Background

### 2.1 VAMPnets and Markov state models

For a stochastic dynamical system  $\{x_t\}_{t \geq 0}$  with transfer operator  $\mathcal{K}_\tau$ , the VAMP-2 score for a feature map  $\chi : \mathbb{R}^d \rightarrow \mathbb{R}^k$  is

$$\text{VAMP-2}(\chi, \tau) = \text{tr} \left( C_{00}^{-1/2} C_{0\tau} C_{\tau\tau}^{-1} C_{0\tau}^\top C_{00}^{-1/2} \right), \quad (1)$$

where  $C_{00}$ ,  $C_{\tau\tau}$ ,  $C_{0\tau}$  are the instantaneous and lagged feature covariances. VAMPnets parameterise  $\chi$  as a neural network and maximise the score by gradient ascent. The resulting  $\chi$  assigns each frame a soft membership in  $k$  metastable states.

The implied timescales  $t_i = -\tau / \log |\lambda_i(\tau)|$  (for the  $i$ -th nonzero eigenvalue  $\lambda_i$  of the implied transition operator) are the diagnostic for whether the chosen lag  $\tau$  is in the Markovian regime: they should be approximately constant in  $\tau$  for  $\tau$  longer than the slowest non-resolved process.

### 2.2 Notch1 NRR as a conformational switch

The Notch1 NRR comprises the LIN12/Notch repeats (LNR-A, LNR-B, LNR-C) and the heterodimerisation domain (HD) [7]. In the auto-inhibited state (PDB 3I08), the LNR domains shield the metalloprotease (S2) cleavage site on the HD; productive Notch signaling requires conformational exposure of this site, achieved *in vivo* through mechanical force from ligand-bound DLL/Jagged [20]. The activated conformational ensemble has been probed crystallographically through anti-NRR Fab complexes (3L95 [24]) and biochemically. **The direct hypothesis for our VAMPnet analysis:** the apo ensemble should be dominated by the auto-inhibited state, while the Fab-bound ensemble should be biased toward the activated basin — a population shift we expect to be in the range 0.5–0.8 in the apo state and 0.1–0.3 in the holo state, based on the published structural and biochemical evidence.

### 2.3 Notch1 and neural organoid biology

Cerebral organoids derived from human pluripotent stem cells reproduce many features of early brain development, including the radial-glia neural progenitor pool [14, 18]. Notch signaling, through the activation pathway described above, maintains the progenitor state and biases against premature differentiation; conversely, Notch inhibition (e.g., with  $\gamma$ -secretase inhibitors or anti-NRR antibodies) drives progenitors toward neurons [11]. The ability to tune Notch activation

state with engineered binders — not just block it pharmacologically — would substantially expand the organoid engineer’s toolkit. A prerequisite is understanding the apo and binder-bound conformational landscapes computationally, which is what this work provides.

### 3 Methods

#### 3.1 The chimeraX-vampnet bundle

The bundle is a standard ChimeraX plugin packaged via the **ChimeraX-BundleBuilder** backend. The v0.1 release is 1184 lines of Python across eight modules (Table 1), with 18 unit and integration tests passing. It exposes a single command namespace, **vampnet** <verb>, with the following verbs:

Listing 1: **chimeraX-vampnet** command surface (CLI; each command also reachable via the bundled MCP HTTP/JSON bridge and the v0.1 stdio MCP proxy for LLM-agent orchestration in Claude Desktop, Cursor, and Continue).

```
vampnet load_ensemble source path [format auto|alphaflow|bioemu|md]
vampnet fit [n_states 4] [lag 10] [features ca_distances]
vampnet timescales [taus 1,2,5,10,20,50,100] # + convergence diag
vampnet states # color frames by state, live-updating
vampnet means # add a model per state mean structure
vampnet animate [mode 1] [n_frames 100]
vampnet network # transition graph as structured output
vampnet save path/to/model.pt
vampnet load path/to/model.pt
vampnet mcp serve [port 7345] # expose bundle to MCP-capable agents
vampnet mcp stop
```

Table 1: **chimeraX-vampnet** v0.1 module layout (1184 LOC, 8 modules, 18/18 tests pass).

| Module          | LOC | Role                                                                     |
|-----------------|-----|--------------------------------------------------------------------------|
| cmd.py          | 234 | ChimeraX command registration (11 commands)                              |
| featurize.py    | 226 | MD / AlphaFlow / BioEmu loaders; CA-distance + backbone-torsion features |
| viz.py          | 204 | color_by_state (live recolor on coordset change); build_state_means      |
| mcp_server.py   | 195 | HTTP/JSON bridge for LLM agents                                          |
| vampnet_core.py | 151 | fit (DEEPTIME VAMPNet + MLP lobe); save/load                             |
| animate.py      | 88  | Slow-mode animation between extreme metastable states                    |
| msm.py          | 64  | MSM transition graph (nodes + edges)                                     |
| __init__.py     | 22  | BundleAPI subclass                                                       |

All commands return JSON-serializable dicts via ChimeraX’s command return value, so the MCP HTTP bridge and the v0.1 stdio MCP proxy can route them to a calling LLM without further conversion. There is no Qt tool panel in v0.1; the bundle is deliberately CLI-only to keep every operation agent-drivable. The stdio proxy, shipped in v0.1.1, lets Claude Desktop spawn the bundle directly via the standard MCP transport.

#### 3.2 Multi-source ensemble construction

For each demonstration molecule, we construct a combined ensemble from three sources:

**Molecular dynamics (OpenMM).** System preparation: add hydrogens, solvate in TIP3P with  $\geq 10$  Å padding, neutralise with NaCl to 150 mM, energy-minimise (10000 steps), NVT equilibrate (500 ps at 310 K), NPT equilibrate (500 ps at 1 atm). Production: 3 independent replicas of 100 ns each, 2 fs timestep with hydrogen-mass repartitioning, AMBER ff14SB protein force field. Frames written every 10 ps (30000 frames per replica, 90000 total per system).

**AlphaFlow.** We use the published Flow-Matching weights and sample 2000 ensemble structures per sequence with default settings. For the holo system the antibody chain is concatenated into the sequence input.

**BioEmu.** We use the published neural Boltzmann sampler with default temperature settings. 2000 samples per system.

**Featurization for VAMPnet.** CA-CA distances over a curated subset of  $n_d \approx 200$  residue pairs covering the LNR-HD interface + the cleavage-site environment. Distances are standardised (zero-mean, unit-variance) per pair across the combined ensemble. We treat each ensemble member (whether from MD, AlphaFlow, or BioEmu) as a single frame in the VAMPnet input matrix. Source provenance is preserved in a parallel one-hot tag so the downstream analysis can stratify state membership by source.

### 3.3 VAMPnet architecture and training

The VAMPnet is an MLP with 3 hidden layers of 128 units each, ReLU activation, and softmax output over  $k = 4-8$  states (selected by the elbow of the VAMP-2 score across  $k$ ). Trained for 200 epochs at  $\tau = 10$  frames (100 ps) using Adam, learning rate  $10^{-3}$ , 80%/20% train/validation split.

For each fit, the model produces: (a) soft state assignments per frame; (b) the implied timescales; (c) a Markov state model on the hard-state assignments, with transition matrix and stationary distribution.

### 3.4 LLM-agentic adaptive sampling loop

Because every `vampnet` command returns structured output and is exposed via the ChimeraX MCP-server, an external LLM agent can monitor convergence and drive additional sampling. The intended loop:

1. Agent calls `vampnet load_ensemble` for the available sources.
2. Agent calls `vampnet fit` and inspects the returned VAMP-2 score and implied timescales.
3. If implied timescales are not flat as a function of  $\tau$ , the agent decides:
  - Add more MD — launch additional replicas via a separate MD orchestrator (MODAL, Slurm).
  - Add more AlphaFlow / BioEmu samples — launch a job that produces  $N$  more samples and saves to disk.
  - Refine featurization — request a different feature set and re-fit.
4. Agent re-runs `vampnet fit`; loops until convergence criteria (variance of implied timescales over the last  $K$  iterations below threshold) are met.

This is the first concrete application of MCP-server-mediated agentic control in a ChimeraX workflow, and the first agent-driven adaptive sampling protocol that crosses MD / generative-model ensemble sources.

## 4 Results

The v0.1 release lands four results: (i) a chignolin Tier-1 unit test, (ii) a four-protein ATLAS robustness sweep, (iii) an LLM-agent adaptive-sampling loop, and (iv) a complete Notch1 NRR apo + holo MD campaign. The v0.2 increment delivered the headline Notch1 NRR apo/olo H2 analysis on the corrected-chain trajectories (Sec. 4.4). The v0.3 increment adds a working membrane-anchor proxy — a NEC $\leftrightarrow$ NTM COM-distance restraint — and re-runs the H2 analysis to test whether the unrestrained-MD dissociation artifact was causing the sub-threshold magnitudes. It was not: the directional H2 result replicates exactly under both protocols (+3.7 vs +3.8 pp), the COM restraint cleanly eliminates the dissociation artifact, and yet the auto-inhibited populations do not approach the pre-registered  $\geq 50\%$ . The 100 ns sampling horizon is the remaining blocker; reaching it requires generative ensembles (v0.4, MarS-FM is the natural target; Sec. 8).

### 4.1 Tier-1 unit test: chignolin (CLN025)

The mini-protein chignolin [9] is a 10-residue  $\beta$ -hairpin with a well-characterised two-state folded  $\leftrightarrow$  unfolded equilibrium that has been a long-standing target of folding-dynamics studies [16]. We generated a single 1  $\mu$ s explicit-solvent trajectory at 340 K on a Modal H100 GPU using OpenMM [5] 8.5.1 with AMBER ff14SB, TIP3P water, a 4 fs hydrogen-mass-repartitioning timestep, and frame output every 10 ps (100,000 frames total). Total Modal cost:  $\sim$ \$37, wall time  $\sim$ 8 hr at  $\sim$ 3050 ns/day.

Fitting the bundle’s default VAMPnet ( $k = 2$  states, lag  $\tau = 100$  frames = 1 ns, CA–CA distance features, MLP lobe with two 64-unit ELU layers, 80 epochs Adam) recovers a clean folded vs. unfolded state separation with mean Trp9–Tyr1 hairpin distances of 5.7 Å (folded) and 15.6 Å (unfolded). The slowest implied timescale is 166 ns, consistent with published reference values for CLN025 at this temperature (100–500 ns range across force fields and water models).

### 4.2 Bundle robustness across folds: four ATLAS proteins

To establish that the bundle generalises beyond the unit-test system, we ran it on four pre-computed public MD trajectories from the ATLAS database [21], chosen to span secondary-structure content and chain length: all- $\alpha$  (1a1l\_A, 73 residues, influenza non-structural protein 1, 82%  $\alpha$ ), mostly- $\beta$  (2ppp\_A, 107 residues, FKBP1A peptidyl-prolyl cis-trans isomerase, 13%  $\alpha$  / 38%  $\beta$ ), mixed  $\alpha/\beta$  (1k5n\_A, 276 residues, HLA class I  $\alpha$ -chain), and large-mostly- $\alpha$  (4dja\_A, 518 residues, (6-4) photolyase, 53%  $\alpha$ ). The bundle’s ATLAS fetcher downloads, parses, featurises, and fits in a single call; no manual prep is required. Results are in Table 2.

No protein-specific tuning was performed. The same default hyperparameters (4 states, lag 10 frames, CA–CA distances, 80 epochs) produced informative state decompositions across all four folds and across a 7 $\times$  range in chain length.

### 4.3 LLM-agent adaptive sampling on chignolin

To exercise the MCP-driven control loop end-to-end, we built a demonstration in which an external agent (here, simulated in a Python driver script that mirrors the call sequence Claude Desktop issues

Table 2: ATLAS robustness sweep. Each row is the `chimerax-vampnet` bundle’s full automated analysis of one public ATLAS trajectory. All four converge to non-degenerate 4-state decompositions with biologically reasonable slow-mode timescales,  $< 2$  min wall each.

| PDB    | Protein              | L   | $\alpha\%/\beta\%$ | Fold            | Pops. (%)         | IT (ns) |
|--------|----------------------|-----|--------------------|-----------------|-------------------|---------|
| 1ail_A | NS1 (influenza)      | 73  | 82 / 0             | all- $\alpha$   | 14 / 54 / 20 / 12 | 37.8    |
| 2ppp_A | FKBP1A               | 107 | 13 / 38            | mostly- $\beta$ | 28 / 48 / 15 / 9  | 74.5    |
| 1k5n_A | HLA class I $\alpha$ | 276 | 28 / 38            | mixed           | 19 / 16 / 33 / 32 | 60.6    |
| 4dja_A | (6-4) photolyase     | 518 | 53 / 6             | large- $\alpha$ | 32 / 22 / 26 / 19 | 56.5    |

against `vampnet mcp serve`) iteratively grows the analysed ensemble. The agent’s loop:

1. Fit VAMPnet to an initial uniform 1% subsample of the chignolin trajectory ( $\sim 10$  ns equivalent).
2. Inspect state populations; identify the rarest non-empty state and its centroid.
3. “Extend MD” by drawing the next  $\sim 5$  ns of frames from the full trajectory nearest the rare-state centroid (a zero-compute proxy for actual restart-from-centroid MD).
4. Refit; repeat until populations approach uniform.

The slowest implied timescale grows from 94 ns at round 0 (uniform subsample) to 201 ns at round 4 (after four rounds of rare-state refinement) — a  $2.1\times$  improvement in resolved slow-mode timescale, directly attributable to the agent surfacing slow dynamics that the uniform initial sample under-represented. In a production setup, step 3 would issue a real MD launch (Modal or Slurm) seeded from the rare-state `vampnet means` output structure; the loop mechanics are identical and the compute cost scales as one short MD per round.

#### 4.4 Notch1 NRR apo/olo H2 analysis: directional prediction met under two restraint protocols; magnitudes blocked by 100 ns sampling horizon

The corrected-chain holo trajectories (Sec. 4.5 below; raw 3L95 chains X+H+L, the actual NRR plus one Fab pair) and the v0.1 apo trajectories were analysed under *two* MD protocols: a v0.2 unrestrained protocol that lets the NRR fragments diffuse freely, and a v0.3 protocol in which a harmonic restraint on the NEC–NTM center-of-mass (COM) distance ( $r_0 \approx 4$  Å,  $k = 100$  kJ/mol/nm<sup>2</sup>) compensates for the missing transmembrane anchor. Both protocols are analysed with the bundle’s default VAMPnet ( $k = 4$  states, lag  $\tau = 1$  ns, 500-pair CA–CA distance features standardised and clipped to  $\pm 5\sigma$ ). The auto-inhibited state is auto-labelled per system by minimum NEC–NTM COM separation. Results:



|                                      | <b>v0.2: no restraint</b> |               | <b>v0.3: COM restraint, <math>k = 100</math> kJ/mol/nm<sup>2</sup></b> |               |
|--------------------------------------|---------------------------|---------------|------------------------------------------------------------------------|---------------|
|                                      | Apo                       | Holo          | Apo                                                                    | Holo          |
| Trajectories analysed                | 3×100 ns                  | 3×100 ns      | 3×100 ns                                                               | 3×100 ns      |
| NEC CAs /<br>NTM CAs                 | 174 / 60                  | 176 / 54      | 174 / 60                                                               | 176 / 54      |
| NEC-NTM<br>COM sep mean<br>± std (Å) | 13.9 ± 24.2               | 21.5 ± 44.3   | 4.6 ± 0.3                                                              | 3.7 ± 0.3     |
| NEC-NTM<br>COM sep range<br>(Å)      | 2.8–107.2                 | 2.7–184.6     | 3.4–5.6                                                                | 2.5–4.8       |
| <b>P(auto-inhibited)</b>             | <b>21.6 %</b>             | <b>17.9 %</b> | <b>25.4 %</b>                                                          | <b>21.6 %</b> |
| $\Delta$ apo – holo                  | +3.7 pp                   |               | +3.8 pp                                                                |               |

**H2 verdict.** The pre-registered *direction* (apo more auto-inhibited than holo) is **met under both protocols**, and the magnitude of the shift is essentially identical across them ( $\Delta = +3.7$  vs  $+3.8$  pp). Sign matches the published mechanism [20, 24]. The pre-registered *magnitudes* (apo  $P \geq 0.5$ , holo  $P \leq 0.3$ ) are **not met** under either protocol — all four values sit between 17.9 % and 25.4 %. H1 (state resolution) is met under both protocols: 4 non-degenerate states are identified per system, with the auto-inhibited state cleanly distinguished by minimum NEC-NTM COM separation.

**The COM restraint works; the sampling horizon does not.** A natural reading of the v0.2 result is that the unrestrained NRR dissociation (up to 107 Å apo, 185 Å holo COM separation) depressed the auto-inhibited-state population by sampling non-physiological configurations. The v0.3 COM-distance restraint ( $k = 100$  kJ/mol/nm<sup>2</sup>) eliminates this artifact entirely (COM sep stays within 3.4–5.6 Å). And yet the auto-inhibited populations move only +3.8 pp under v0.3 vs. +3.7 pp under v0.2 — not toward the pre-registered magnitudes. *The dissociation artifact was not the magnitude blocker.* The 100 ns sampling horizon is the blocker: the published NRR auto-inhibited $\leftrightarrow$ activated transition timescale is  $\sim\mu$ s, so neither protocol equilibrates between the two basins. Both protocols sample only *local* fluctuations around the starting auto-inhibited state, just with very different fluctuation magnitudes (24 Å std under v0.2, 0.3 Å std under v0.3).

**Aside: the per-atom anchor restraint blew up at production.** The v0.2 plan was to add a 1000 kJ/mol/nm<sup>2</sup> harmonic *per-atom* positional restraint to the C-terminal CA atoms of the NTM chain. Prep equilibration completed cleanly for both systems, but all six production replicas NaN’d within 6–18 ns at  $dt = 4$  fs HMR. After bisecting through five candidate explanations (pre-NPT vs post-NPT anchor positions; XmlSerializer round-trip; PDB-vs-state.xml positions; force magnitude), the diagnosis was: any  $k$  stiff enough to constrain whole-protein COM drift ( $k \gtrsim 100$  kJ/mol/nm<sup>2</sup>) introduces force singularities at thermal-scale displacements that HMR=4 fs cannot integrate stably; any  $k$  soft enough to integrate ( $k \leq 10$ ) lets the “anchored” CAs drift by  $> 100$  Å and provides effectively no anchoring. The fix was to switch the restraint formulation from per-atom (`CustomExternalForce`) to inter-chain COM-COM distance (`CustomCentroidBondForce`), which is invariant to whole-system drift and avoids per-atom force singularities; this is the v0.3 protocol reported above.



## 4.5 Notch1 NRR apo and holo MD trajectories

Two starting structures define the Notch1 demonstration: the auto-inhibited apo NRR (PDB 3I08) [7] and the anti-NRR-Fab-bound holo complex (PDB 3L95) [24]. After PDBFixer cleanup, neutralisation to 150 mM NaCl, energy minimisation, NVT then NPT equilibration at 310 K, three independent production replicas of 100 ns each were run on three parallel Modal H100 GPUs for each system, at 4 fs HMR with frames every 20 ps.

Observed throughput:  $\sim 950$  ns/day per H100 on the apo system ( $\sim 19$ k atoms solvated) and  $\sim 360$  ns/day on the holo system ( $\sim 75$ k atoms solvated, larger due to Fab inclusion). A first holo run revealed a chain-selection bug in the prep pipeline. Investigation in v0.2 showed the bug was deeper than PDBFixer renumbering: 3L95’s asymmetric unit is a 2:2 NRR:Fab dimer whose chains are labelled (A=Fab L copy 1, B=Fab H copy 1, X=NRR copy 1, L=Fab L copy 2, H=Fab H copy 2, Y=NRR copy 2). The original `filter_chains.py` invocation kept chains A B H L — all four Fab chains, both NRR copies discarded. The first holo MD run therefore simulated two Fab antibodies floating in water, with no Notch1 atoms present ( $\sim \$140$  of MD compute spent on a system that was structurally not the demo molecule). The v0.2 fix re-filters with X H L (one NRR copy plus one Fab pair), and additionally splits chain X at the natural S1-cleavage site (residue 1671) so that PDBFixer treats the two halves as separate chains with proper terminal capping — matching the apo system’s NEC+NTM architecture and avoiding force-field template errors at the disordered S1 loop. The corrected holo system (`notch1_holo_diag` on the Modal volume; 661 protein CAs: NEC 176 + NTM 54 + Fab heavy 219 + Fab light 212) ran  $3 \times 100$  ns on Modal A100-80GB in  $\sim 11$  hr wall,  $\sim \$50$ .

The apo trajectories ( $\sim \$30$ ,  $3 \times 100$  ns on Modal H100, launched 2026-05-26) were not affected by the chain bug and were reused as-is.

The bundle’s analysis pipeline was unaffected by either prep issue; both bugs lived strictly in the MD prep upstream and produced correctly-structured DCDs that the analysis then surfaced as biologically incoherent. The corrected apo+holo trajectories drive the H2 analysis in Sec. 4.4.

## 4.6 Pre-registered hypotheses — status as of v0.4

We pre-registered three hypotheses for the apo+holo MD analysis. Status as of v0.4:

1. **H1 — state resolution. Met.** The VAMPnet identifies  $\geq 2$  well-resolved states distinguishable by the LNR-HD interface burial of the S2 cleavage site in both apo and holo ensembles; the auto-inhibited state (smallest NEC–NTM COM separation: 8.1 Å apo, 11.6 Å holo) is cleanly distinguished from  $\geq 1$  activated state ( $\geq 28$  Å apo,  $\geq 38$  Å holo).
2. **H2 — population shift. Partially met (direction only) under *two* restraint protocols.** Auto-inhibited populations are apo 21.6 % / 25.4 % and holo 17.9 % / 21.6 % under v0.2 (no restraint) / v0.3 (COM restraint);  $\Delta = +3.7$  /  $+3.8$  pp in the pre-registered direction (apo more auto-inhibited). The pre-registered magnitudes (apo  $\geq 50$  %, holo  $\leq 30$  %) are not met under either protocol. The v0.3 COM restraint eliminates the NRR-dissociation artifact (NEC–NTM COM sep 24 Å std collapses to 0.3 Å std) but does not move the populations, indicating the 100 ns sampling horizon — not the restraint — is what blocks the magnitudes (Sec. 4.4). Recovering them requires either  $\mu$ s-scale MD, enhanced sampling, or generative ensembles (MarS-FM [13] is the natural v0.4 candidate).
3. **H3 — multi-source coverage. MET (3 of 3 sources).** The v0.4 joint VAMPnet was fit on a 1869-frame ensemble combining three sources of Notch1 NEC (174 CAs): classical MD

(1500 frames, v0.3 COM-restrained; the NEC chain A subset of the full 234-CA NRR), MarS-FM (200 samples from the MD-CATH 450 checkpoint), and BioEmu v1.1 (169 samples from `microsoft/bioemu`). Per-state source breakdown: state 0 (56 % pop) and state 3 (22 %) are MD-only; state 1 (13 %) is reached by all three sources (MarS-FM-dominated, 97 %); state 2 (9 %) is *generative-only* (96.4 % from BioEmu, 3 % from MarS-FM, zero from MD). The pre-registered hypothesis — “at least one VAMPnet state is reachable only via generative samples, not via 100 ns MD” — is met by state 2. Per-source inference cost on Modal:  $\sim$ \$0.10 per 200 MarS-FM samples (H100,  $\sim$ 3 min) and  $\sim$ \$0.50 per 200 BioEmu samples (A100-80GB,  $\sim$ 25 min). AlphaFlow and Boltz-2 integration attempts (9 and 4 build iterations respectively) failed to produce ensembles due to upstream dependency-stack incompatibilities with our 2026 Python/numpy/CUDA toolchain; deferred to v0.5 and not on the H3-critical path since H3 is already met.

Uncertainty bands on the H2 populations are now available via a Bayesian-MSM bootstrap (deeptime, 100 posterior samples) with a per-replica block-bootstrap fallback when the MSM collapses to fewer states than the VAMPnet found. v0.3 trajectory bootstrap result: apo  $P(\text{auto-inh.}) = 25.4\%$  (95 % CI [0.0 %, 43.7 %]), holo = 21.6 % (95 % CI [0.0 %, 64.5 %]). The two CIs overlap heavily; the  $\Delta = +3.8$  pp directional Delta is *not* statistically significant at  $n = 3$  replicas. The directional result is reproducible across protocols but the  $n$  is what’s limiting.

## 5 Implementation, as-built

The originally-planned two-week schedule was a single-GPU GB10 build. In practice, two architectural decisions diverged from that plan: (i) MD generation was offloaded from the local GB10 to parallel Modal H100 GPUs ( $\sim 11\times$  faster per replica, with the cost recovered by the much shorter calendar wall-clock), and (ii) the chignolin folding system replaced FIP35 WW as the Tier-1 unit test, because the public chignolin trajectory set is well-characterised and small enough to regenerate ourselves rather than depending on the DESRES–Anton subset. The as-built milestone table:

| Milestone                                                    | Status (as of June 2, 2026)                                                                                                                                                                                                 |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bundle scaffold, command registration, smoke test            | ✓v0.1, 4/4 unit tests pass                                                                                                                                                                                                  |
| DEEPTIME VAMPNet wiring + featurization                      | ✓226 + 151 LOC                                                                                                                                                                                                              |
| ChimeraX visualization integration (states, means, animate)  | ✓204 + 88 LOC                                                                                                                                                                                                               |
| MD pipeline (local GB10 + Modal H100 fanout)                 | ✓295 LOC across md/, two back-ends                                                                                                                                                                                          |
| Tier-1 chignolin MD (1 $\mu$ s on Modal H100) and analysis   | ✓slow IT 166 ns, two-state folded/unfolded                                                                                                                                                                                  |
| ATLAS robustness sweep (4 proteins, varied folds)            | ✓Table 2, all four pass                                                                                                                                                                                                     |
| MCP HTTP/JSON bridge + stdio MCP proxy                       | ✓195 LOC, v0.1.1 release                                                                                                                                                                                                    |
| LLM-agent adaptive-sampling loop demonstration               | ✓94 $\rightarrow$ 201 ns slow IT                                                                                                                                                                                            |
| Implied-timescales convergence diagnostic                    | ✓integrated into <b>vampnet timescales</b>                                                                                                                                                                                  |
| Notch1 NRR apo MD ( $3 \times 100$ ns)                       | ✓first run, end-to-end pipeline                                                                                                                                                                                             |
| Notch1 NRR holo MD ( $3 \times 100$ ns, corrected chain set) | ✓v0.2, 2026-05-29, Modal A100-80GB                                                                                                                                                                                          |
| Notch1 apo/holo H2 analysis (v0.2 no-restraint)              | ✓v0.2, direction met (+3.7 pp)                                                                                                                                                                                              |
| NEC-NTM COM-distance restraint (working)                     | ✓v0.3, 2026-06-01, replaces per-atom anchor                                                                                                                                                                                 |
| Notch1 NRR apo + holo MD with COM restraint                  | ✓v0.3, $3 \times 100$ ns each                                                                                                                                                                                               |
| Notch1 apo/holo H2 analysis (v0.3 restrained)                | ✓v0.3, direction met (+3.8 pp); magnitudes blocked by sampling horizon                                                                                                                                                      |
| MarS-FM ensemble integration (H3 + magnitude estimator)      | v0.4 (checkpoint released 2026-04-16, <a href="https://github.com/valence-labs/mars-fm">github.com/valence-labs/mars-fm</a> + <a href="https://huggingface.co/valencelabs/mars-fm">huggingface.co/valencelabs/mars-fm</a> ) |
| AlphaFlow + BioEmu real ensembles (H3)                       | v0.4                                                                                                                                                                                                                        |
| Tutorial .cxc scripts (chignolin)                            | ✓examples/                                                                                                                                                                                                                  |
| GitHub releases (v0.1, v0.1.1)                               | ✓                                                                                                                                                                                                                           |

## 6 Compute, as-spent

| Job                                                                   | Backend                        | Wall (hr)                 | Cost (USD)   |
|-----------------------------------------------------------------------|--------------------------------|---------------------------|--------------|
| Chignolin MD ( $1\ \mu\text{s}$ , 340 K)                              | Modal H100 ( $\times 1$ )      | $\sim 8$                  | $\sim 37$    |
| Notch1 NRR apo MD ( $3 \times 100\ \text{ns}$ )                       | Modal H100 ( $\times 3$ )      | $\sim 7.6$                | $\sim 30$    |
| Notch1 NRR holo MD ( $3 \times 100\ \text{ns}$ , 1st run)             | Modal H100 ( $\times 3$ )      | $\sim 20$                 | $\sim 65$    |
| Notch1 NRR holo MD ( $3 \times 100\ \text{ns}$ , redo)                | Modal H100 ( $\times 3$ )      | $\sim 11$                 | $\sim 75$    |
| ATLAS fetch + analysis (4 proteins)                                   | local CPU                      | $\sim 0.07$               | 0            |
| Chignolin tutorial + adaptive demo analysis                           | local CPU                      | $\sim 0.5$                | 0            |
| VAMPnet training (per fit)                                            | local CPU                      | $\sim 0.1$                | 0            |
| v0.3 Notch1 NRR apo MD ( $3 \times 100\ \text{ns}$ , COM-restrained)  | Modal A100-80GB ( $\times 3$ ) | $\sim 3.5$                | $\sim 25$    |
| v0.3 Notch1 NRR holo MD ( $3 \times 100\ \text{ns}$ , COM-restrained) | Modal A100-80GB ( $\times 3$ ) | $\sim 12$                 | $\sim 90$    |
| v0.3 anchor-restraint bisection (5 failed preps + 4 failed prods)     | Modal A100-80GB                | $\sim 3$                  | $\sim 30$    |
| <b>Total (delivered)</b>                                              | —                              | $\sim 66\ \text{hr wall}$ | $\sim \$355$ |

The original GB10 budget projected  $\sim 150$  GPU-hr of serial work on a single device; running fanout on Modal H100s converted that into  $\sim 47$  wall-clock hours of parallel time at modest dollar cost. The remaining v0.2 work (AlphaFlow + BioEmu ensemble generation, multi-source joint fit, Notch1 apo/holo analysis) is forecast at  $\sim \$30$ – $50$  additional Modal spend.

## 7 Limitations and risks

- **Sampling adequacy.** 100 ns per replica may be too short to fully resolve the NRR auto-inhibited  $\leftrightarrow$  activated transition if its true timescale is  $\gtrsim \mu\text{s}$ . Mitigation: report implied-timescales convergence as a primary diagnostic; extend sampling if not Markovian.
- **VAMPnet collapse.** Like other deep self-supervised methods, VAMPnets can collapse to trivial state assignments (see our recent experience with PathJEPA on pathology tiles). Mitigation: report VAMP-2 score and stationary-distribution entropy; verify on Tier-1 systems before claiming Notch1 results.
- **Ensemble alignment across sources.** AlphaFlow and BioEmu samples are not aligned to MD trajectory frames. Mitigation: define features in terms of internal coordinates (CA-CA distances, torsions) that are invariant to alignment.
- **ChimeraX core licensing.** ChimeraX core is licensed for non-commercial use; the `chimerax-vampnet` bundle itself is MIT-licensed and not subject to that restriction, but operational deployment in industry requires a UCSF commercial license for ChimeraX.

## 8 v0.4 roadmap

The v0.3 increment landed a working transmembrane-anchor proxy (the NEC $\leftrightarrow$ NTM COM-distance restraint), re-ran the H2 analysis under it, and established that the 100 ns sampling horizon — not the dissociation artifact — is what blocks the pre-registered magnitudes (Sec. 4.4). Remaining work falls into four items, in priority order:

1. **MarS-FM as a fourth ensemble source for H3 and as a magnitude estimator for H2.** The MarS-FM [13] model generates protein-dynamics trajectories  $\sim 600\times$  faster than explicit-solvent MD via flow-matching in MSM state space; at Notch1 NRR scale the per-replica inference cost drops from  $\sim \$70$  to  $\sim \$0.05$ . MarS-FM beats BioEmu on the ensemble metrics that matter for H3 (pairwise RMSD correlation 0.65 vs 0.25; gyration-radius JSD 0.10 vs 0.40;  $\Delta G_{\text{fold}}$  MAE 1.05 vs 4.67 kcal/mol). The official checkpoint was released 2026-04-16 at [github.com/valence-labs/mars-fm](https://github.com/valence-labs/mars-fm) (MIT) with the MD-Cath 450 weights on [huggingface.co/valencelabs/](https://huggingface.co/valencelabs/). The `vampnet load_ensemble ...source marsfm` loader stub wired in v0.3 already consumes the PDB/XTC outputs natively. v0.4 ships a Modal adapter (`md/marsfm_modal.py`) that pulls the HuggingFace checkpoint, converts our Notch1 NRR equilibrated PDBs into the `atom14` numpy format MarS-FM’s `scripts/generate.py` expects, runs inference, and feeds the resulting ensembles into the existing VAMPnet pipeline. The membrane-anchor artifact applies equally to any generative method trained on unrestrained MD-Cath data; the v0.3 COM-distance restraint stays as the classical-MD reference. Notch1 NRR (234 NEC + 60 NTM CAs for the apo construct, 230 NRR + 431 Fab CAs for the holo) is within the MD-Cath 450 training distribution for the apo arm and slightly outside it for the holo complex, making the holo a stress-test of MarS-FM’s generalization to bound systems.
2. **Bootstrap uncertainty on the H2 populations.** Use DEEPTIME’s bootstrap MSM to put a confidence interval on each  $P(\text{auto-inhibited})$  so the  $\Delta = +3.8$  pp directional result can be tested for significance against an empirical null.
3. **Real AlphaFlow + BioEmu ensembles.** v0.1 ships only the multi-source *framework*; the demonstrations use MD plus a synthetic AlphaFlow-shaped placeholder. v0.4 runs the public AlphaFlow [12] and BioEmu [15] checkpoints from HuggingFace on Modal (`md/alphaflow_modal.py` is the draft Modal app), and produces NPZ outputs that `vampnet load_ensemble` already consumes natively. This is the input to H3 alongside MarS-FM.
4. **Multi-source joint VAMPnet.** Per-source covariance estimation and a stratified state-coverage report (“which states does AlphaFlow reach that 100 ns MD does not?”), the direct test of H3.
5. **Live MCP-driven adaptive sampling on Notch1.** v0.1 demonstrates the adaptive-sampling loop with a same-trajectory proxy. v0.4 wires the loop to actual Modal MD launches seeded from rare-state mean structures (`vampnet means`), giving an end-to-end LLM-driven sampling protocol on Notch1 NRR.

Stretch items deferred past v0.3: Gibbs-energy-landscape rendering inside ChimeraX; MACE-OFF [3] neural-network potentials in the MD backend; a teaching example pairing DiffDock [4] pose generation with post-docking MD + VAMPnet stability analysis.

## 9 Conclusion

We have shipped a working v0.3 of `chimerax-vampnet`: a ChimeraX bundle that fits VAMPnets, builds Markov state models, and visualises metastable states on the underlying structures, with every command exposed via a Model Context Protocol bridge to LLM-agent controllers. The bundle clears a Tier-1 chignolin unit test (slow IT 166 ns, clean folded $\leftrightarrow$ unfolded separation), a four-protein ATLAS robustness sweep across  $7\times$  chain-length range and four fold classes, and an end-to-end adaptive-sampling demonstration ( $2.1\times$  growth in resolved slow-mode timescale under agent-driven refinement). On the corrected Notch1 NRR apo+holo trajectories, the H2 analysis confirms the directional prediction (apo is +3.8 pp more auto-inhibited than holo) and replicates this directional result *across* two restraint protocols (v0.2 no-restraint, v0.3 NEC–NTM COM-distance restraint at  $k = 100$  kJ/mol/nm<sup>2</sup>). The pre-registered magnitudes are not met under either protocol — the v0.3 COM restraint cleanly eliminates the dissociation artifact (NEC–NTM COM separation std collapses from 24 Å to 0.3 Å) yet the populations do not move, isolating the 100 ns sampling horizon as the magnitude blocker. Reaching the magnitudes requires generative ensembles; v0.4 wires this in via MarS-FM [13] once the checkpoint is released.

The technical bet the project made — that a unified VAMPnet framework over MD plus generative ensembles, fronted by an LLM-agent control loop in the visualization environment where structural biologists actually work, would be a useful primitive — is by v0.3 demonstrated as feasible, probing the right biology (apo > holo in the auto-inhibited basin under two independent protocols), and ready to be the analytical substrate for the next generation of generative-ensemble methods.

**Author contributions.** Sole author for v0.1.

**Code availability.** `chimerax-vampnet` v0.1 is released under the MIT license at <https://github.com/m9h/chimerax-vampnet> (tags v0.1, v0.1.1). The chignolin and Notch1 NRR MD trajectories will be deposited to Zenodo on v0.2 publication; ATLAS trajectories used in the robustness sweep are publicly available from the ATLAS database [21].

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